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Belt tensioner with integrated idler pulley for an endless drive belt of an engine

Abstract

A belt tensioner is disclosed that has a spring case defining a receptacle for a torsion spring and having an arm extending radially outward and axially rearward therefrom and terminating with a bore configured for fastening the spring case directly to an engine. The belt tensioner has a support base non-rotationally connected to the spring case through a pivot tube. The pivot tube defines a first axis of rotation, and a pulley mount of the support base defines a second axis of rotation. The support base includes at least one fastener bore for connection to the engine. An arm is coupled to the support base for rotation about the first axis in response to a spring force applied by a torsion spring seated within the spring case and has a free end that defines a pulley mount that defines a third axis of rotation.

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Background/Summary

RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application No. 63/480,889, filed, Jan. 20, 2023 and U.S. Provisional Application No. 63/482,409, filed Jan. 31, 2023, which are both incorporated herein by reference in their entirety.

TECHNICAL FIELD

(1) The present invention relates generally to belt tensioners and more particularly to a belt tensioner with an integrated idler pulley, where the belt tensioner has a spring case with a radially extending anchor terminating with a mounting hole configured for direct fastening to an engine mount.

BACKGROUND

(2) Front end accessory drive systems of engines include an endless belt operatively driven by a crank shaft and various pulleys some of which may drive accessories. A belt tensioner is typically present in such drive systems. One example drive system and an example belt tensioner are disclosed in U.S. application Ser. No. 17/034,676, which is included herein as FIGS. **1**A and **1**B. This belt tensioner has three mounting holes in the support bracket and a pivot tube of the belt tensioner offset from the mounting holes. FIG. **1**B includes the force path for the hub load applied by the belt to the pulley as a dashed-dotted line. It has been determined that the pivot tube can pitch in the hub load direction when load is applied, which can cause the spring case to roll resulting in bending forces acting on the support bracket. While the belt tensioner is effective and meets safety standards, it is desirable to improve the same by reducing or eliminating the bending forces experienced by the belt tensioner, especially in the support bracket. Thus, there is a need for an improved belt tensioner.

SUMMARY

(3) In all aspects, a belt tensioner is described that has a spring case defining a receptacle for a torsion spring that has an arm extending radially outward and axially rearward therefrom, which terminates with a bore configured for fastening the spring case directly to an engine. The belt tensioner has a support base non-rotationally connected to the spring case by a pivot tube that defines a first axis of rotation. The support base has a pulley mount that defines a second axis of rotation and at least one bore configured for fastening the support base to the engine also. The belt tensioner has an arm having its first end coupled to the pivot tube for rotation about the first axis relative to the spring case and support base and having a second end that defines a pulley mount that defines a third axis of rotation. The belt tensioner has a torsion spring having a first end operatively coupled to the support base and a second end operative coupled to the arm such that the torsion spring biases the arm in a belt engaging direction relative to the support base.

(4) In all aspects, the pivot tube is positioned outside an area defined by the bores. The pivot tube can have a knurled first end fixedly connected to the spring case and a knurled second end fixedly connected to the support base.

(5) In all aspects, the torsion spring can be a flat wire spring. The first end of the spring is attached to the spring case, and the first end of the arm includes a pivot tube-receiving feature to which the second end of the spring is operatively attached.

(6) In all aspects, the belt tensioner has a first pulley operatively connected to the pulley mount of the support base for rotation about the second axis and a second pulley operatively connected to the second end of the arm for rotation about the third axis.

(7) In all aspects, a spring bushing seated between the arm and the torsion spring, and a damper connected to the arm for rotation therewith and in operational engagement with the support base. In one embodiment, the damper is keyed to the first end of the arm for rotation with the arm. The spring bushing can have a center ring and an outer ring connected to one another by a plurality of

spokes, and has an upper outermost axial flange seated against an exterior surface of the spring case and a lower outermost axial flange seated against an exterior surface of the arm.
(8) In all aspects, the spring case can have an arm travel limiting feature defining a preselected number of degrees of rotation for the arm and the arm has a stop configured to engage the arm travel limiting feature.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1**A is a top plan view of a prior art belt tensioner having an integrated idler pulley.
(2) FIG. **1**B is a longitudinal, cross-sectional view of the belt tensioner portion taken along line A-A in FIG. **1**A.
(3) FIG. **1**C is a front plan view of an internal combustion engine system of the prior art.
(4) FIG. **2** is a top plan view of a first embodiment of a belt tensioner having an integrated idler pulley.
(5) FIG. **3** is a longitudinal, cross-section through the belt tensioner portion of FIG. **2** taken along line **3-3**.
(6) FIG. **4** is an exploded view of the belt tensioner of FIG. **2** from a bottom perspective view.
(7) FIG. **5** is an exploded view of the main body of the belt tensioner of FIG. **2** from a top perspective view.
(8) FIG. **6** is a bottom perspective view of the spring case having seated therein a torsion spring and a spring bushing.
DETAILED DESCRIPTION
(9) The following detailed description will illustrate the general principles of the invention, examples of which are additionally illustrated in the accompanying drawings. In the drawings, like reference numbers indicate identical or functionally similar elements.
(10) Referring to FIGS. **2-5**, a first embodiment of a belt tensioner **100** for providing a predetermined amount of tension upon a belt (labeled in FIG. **2**), often found in a front end accessory drive (FEAD) of an engine system, is disclosed. The belt tensioner **100** is useful in a FEAD such as the transmission belt system exemplified in FIG. **1**C taken from U.S. application Ser. No. 17/034,676 in place of belt tensioner **28**. Herein, referring to FIGS. **2** and **3**, the belt tensioner **100** has a main body **102** that includes an arm **106** pivotally mounted to a support base **108** by a pivot tube **110** defining a first axis (X.sub.1) of rotation about which the arm pivots or rotates and a torsion spring **112** operatively coupled between the arm **106** and the support base **108** that biases the arm in a belt-tensioning direction T. The support base **108** is fixed to the pivot tube **110**, has a pulley mount **120** (best seen in FIGS. **4** and **5**) that defines a second axis (X.sub.2) of rotation, and at least one bore **122** configured for fastening the support base to an engine.
(11) As labeled in FIGS. **4** and **5**, the arm **106** has a first end **130** defining a pivot tube-receiving feature **132** and has a second end **134** that defines a pulley mount **136** that defines a third axis of rotation (X.sub.3). The main body **102** includes a spring case **111** defining a receptacle **113** for the torsion spring **112**. The spring case **111** has an arm **117** extending radially outward and axially rearward therefrom (as best seen in FIGS. **2** and **5**) that terminates with a bore **118** configured for fastening the spring case directly to an engine, i.e., independent of the support base (the bore **118** does not rest upon the support base or align with a bore in the support base). The fastening directly to the engine can be to the cylinder block or to a housing or other structure supported upon the cylinder block. A bolt received in bore **118** passes through the arm **117** and fastens to the engine. The spring case **111** is non-rotationally connected to the support base **108** via the pivot tube **110**, but both have at least one anchor point for direct fastening to the engine. The embodiment of FIGS. **2-5** has two bore **122***a*, **122***b* in the support base **108** and one bore **118** in the arm **117** of the spring

case.

(12) As shown in FIG. **2**, the pivot tube **110** is mounted to the support base **108** at a position that is outside the area defined by the bores **112***a*, **122***b* and **118**. When the fasteners are bolts, this can be described as a position outside of the bolt pattern. As seen in FIGS. **5** and **6**, the second end **144** of the pivot tube **110**, which may be knurled, seats in a mounting bore **124** of the support base **108** and is fixedly attached thereto. The fixed attachment of the pivot tube **110** to the support base **108** can be a radially riveted connection. In this improved belt tensioner **100**, the first end **142** and the second end **144** of the pivot tube **110** are both anchored to the engine. The first end **142** is anchored by its fixed connection to the spring case **111**. The second end **144** is anchored by its fixed connection to the support base **108**. As shown in FIG. **3**, the arm will flex under the hub load, but the force path (the bold dashed lines) is shared between the spring case **111** and the support base **108**. More specifically, the force path divides at the pivot tube **110** and a portion of the force is directed to the first end **142** of the pivot tube **110** fixedly connected to the spring case **111** and from there to the arm **117** of the spring case to its fixed attachment point at mounting bore **118** to the engine. The other portion of the force path is directed to the second end **144** of the pivot tube **110** fixedly connected to the support base **108** to its fixed attachment point (at least) at mounting bore **122***b* to the engine. The result is reduced flexing in the support base **108** because the spring case **111** shares the load back to datum A. As such, the belt tensioner has improved alignment control due to reduced component flexing under the hub load force.

(13) Turning now to FIG. **4**, a first pulley **150** is mounted for rotation to the second end **134** of the arm **106** by a bolt **152** extending through the hub **154** of the pulley **150** and into the pulley mount **136** of the arm **106**. The pulley **150** is preferably journaled to the second end **134** by a roller bearing **154**. A dust cover **158** is coaxially mounted to the pulley **150** to protect the roller bearing **154** from debris and contamination. In the first embodiment, the dust cover is seated over the head of the bolt. In another embodiment, the dust cover may be seated between the head of the bolt **152** and the hub of the pulley **150**. To seal the bottom of the pulley **150** against debris and contamination an annular seal **156**, such as a V-ring, X-ring, or O-ring seal, can be seated in operative engagement with the roller bearing **154** and the arm **106**.

(14) Still referring to FIG. **4**, a second pulley **160**, referred to as an idler pulley, is mounted to the support base **108** at the pulley mount **120** for rotation about the second axis (X.sub.2). The second pulley **160** is mounted thereto by a bolt **162** extending through the hub **164** of the pulley **160** and into the pulley mount **120**. The pulley **160** is preferably journaled to the pulley mount **120** by a roller bearing **164**. A dust cover **168** is coaxially mounted to pulley **160** to protect the roller bearing **164** from debris and contamination. In the first embodiment, the dust cover is seated over the head of the bolt. In another embodiment, the dust cover may be seated between the head of the bolt **162** and the hub of the pulley **160**. To seal the bottom of the second pulley **160** against debris and contamination an annular seal **166**, such as a V-ring, X-ring, or O-ring seal, can be seated in operative engagement with the roller bearing **164** and the support base **108**.

(15) Referring to FIGS. **4** and **5**, the arm **106** has a pivot tube-receiving feature **132** (best seen in FIG. **5**) protruding from the first end **130**. An annular pivot bushing **133** is seated inside the pivot tube-receiving feature **132**. The annular pivot bushing **133** is bounded by an upper sealing member **190** and a lower sealing member **192**, which prevent contaminants and debris from entering the pivot tube-receiving feature **132**. The sealing members **190**, **192** can be a V-ring, X-ring, O-ring seals or any equivalent thereto. The pivot tube-receiving feature **132** includes a spring seat **135** for the first end **114** of the torsion spring **112**. The opposing side of the first end **130** of the arm **106** includes an annular protruding hub **137** that includes either a key or keyway for non-rotational connection to a damper **170**, which has a mating key or keyway **172**.

(16) The damper **170** includes a damper ring **171** keyed to the annular protruding hub **137** of the arm **106**. A washer **176** can be present between the damper ring **171** and a biasing member **174**. The biasing member can be a Bellville spring as shown or any other type of biasing member

suitable for applying an axial force on the damper ring **171** to keep it in frictional contact with the support base **108**. The damper **170** urges the arm towards the spring bushing to take up any clearances as the bushings wear during general operation and component stack up that might occur during assembly. During normal tensioning, when a belt presses against the pulley **150** of the arm **106**, the arm will rotate about the pivot axis X.sub.1 thereby winding the torsion spring **112**. The torsion spring after being wound will apply spring torque against the arm **106** to move, hold, or press the arm and pulley against the belt. When the arm **106** rotates about the pivot axis X.sub.1 in either direction, the frictional contact between the support base **108** and the damper **170** reduces or acts to minimize the rotation of the arm. The damper's operative engagement with the arm **106** and frictional contact with the support base provides frictional symmetric damping.
(17) The arm **106** can include a tool receptacle **180** (FIGS. **2** and **5**). A tool can be inserted to apply torque to the arm to move the pulley **150** away from the belt for example during installation of the belt tensioner **100**, which is referred to in FIG. **2** by arrow W (winding of the torsion spring). Referring now to FIGS. **2** and **5**, the support base **108** includes a bore **182** that can be aligned with a tab **184** of the arm **106** that defines a bore **185**, and once aligned a locking pin (not shown) can be inserted to hold the arm **106** in an install position.
(18) Referring again to FIG. **5**, the spring case **111** has an arm travel limiting feature **138**, which may be referred to as a stop, defining a preselected number of degrees of rotation for the arm **106**. The arm **106** has a protrusion or a second travel limiting feature **139**, which may also be referred to a stop, configured to engage the arm travel limiting feature **138** of the spring case. When these two stops are engaged, the arm is in a free arm position. Additionally, the tab **184** of the arm **106** is configured as a second stop of the arm. The tab **184** can engage a partial rib **109** of the bracket **108** when the arm **106** is in an install position.
(19) Referring again to FIGS. **3-5**, the torsion spring **112** applies a torsional spring force on the arm **106** in the direction shown by arrow T (FIG. **2**) representing a belt engaging or belt tensioning direction, such that the second end **134** of the arm **106** applies a corresponding tension force upon the belt. In the illustrated embodiments, the torsion spring **112** is a flat wire spring having a spring tape **140** interleaved between the coils thereof, but the belt tensioner is not limited thereto. In other embodiments, the torsion spring **112** can be a coil spring, such as a round wire coil spring. The use of spring tape **140** reduces frictional wear of the spring or other negative effects of friction such that spring collapse is reduced. The flat wire torsion spring **112** has an inner spring end **114** attached to the arm **106**, more specifically to the pivot-tube receiving portion **132** and an outer spring end **116** attached to the spring case **111**, more specifically seated in an open notch **119** in an outer wall **115** of the spring case **111** (best seen in FIG. **5**). The inner and outer spring ends **114**, **116** can be bent to define hooks, tangs, etc. to enhance the attachment of the spring to the respective pivot tube and arm. When assembled, a plug **196** can be inserted into the open notch **119** next to the outer spring end **116** of the torsion spring **112** to block contaminants and debris from entering the spring case. The open notch **119** defines a spring abutment feature against which the outer spring end **116** is operatively seated to bias the arm **106** in the belt tensioning direction.
(20) Referring now to FIGS. **3** and **6**, the tensioner includes a spring bushing **198**. The spring bushing **198** is seated between the arm **106** and the torsion spring **112**. The spring bushing **198** has a center ring **200** and an outer ring **202** connected to one another by a plurality of spokes **204**. The spring bushing **198** includes an upper outermost axial flange **206** and a lower outermost axial flange **208**. The upper outermost axial flange **206** is seated against an exterior surface of the arm **106** and the lower axial outermost flange **208** is seated against an exterior surface of the spring case **111**. This bushing thus creates a seal between the arm and spring case to prevent contaminants and debris from entering the main body **102** of the tensioner **100**. This configuration renders the outer ring **202** to be T-shaped when viewed in longitudinal cross-section, as shown in FIG. **3**. Alternately, the outer ring **202** can be H-shaped when viewed in longitudinal cross-section, thereby providing sealing against both the interior surface and the exterior surface of both the spring case and the arm.

The spokes **204** rest against the torsion spring **112**. The surface of the arm facing the pivot bushing can include a plurality of fingers protruding therefrom that are in a spaced arrangement to have one each between immediately neighboring spokes **204**. The spokes **204** and fingers **178** of the arm (labeled in FIG. **5**) can either or both functions to maintain the spring tape **140** between the coils of the torsion spring **112**, i.e., prevent the spring tape **140** from migrating out from between the coils during operation of the tensioner.
(21) The arm **106**, support base **108**, pivot tube **110**, and/or spring case **111** are often made of metallic material such as solid steel, aluminum, or powdered metal and some plastics if of sufficient strength. These parts may be die-cast parts. The torsion spring **112** may be manufactured from steel, but other suitable alternative materials (or combination of materials/components) to construct such components are also contemplated. The bushings are typically a wear resistant plastic. The wear resistant plastic can be, but is not limited to, all polyamides (PA) including 66 nylon, 6 nylon, 11 nylon, 12 nylon, 69 nylon, 612 nylon, and 610 nylon, polyamide 46 nylon; polyethermides (PEI); polysulfones (PSU); polyethersulfones (PES); polyoxymethylenes (POM), or acetals; polyetheretherketones (PEEK); polyphenylene sulfides (PPS); polypthalamides (PPS), or amodels; polyphenylene sulfides (PPO); and amorphous nylons. While the bushings in this embodiment are wear resistant plastic and PTFE coated steel, it is within the scope of the invention to use other suitable bushing materials or bearing structures. The bushing provides a bearing surface for the rotation of elements of the tensioner about the pivot axis.
(22) With reference to FIGS. **4-6**, during assembly, the spring case **111** is mounted in a support of a supporting machine (not shown), the pivot tube **110** can be inserted and fixed to the spring case **111** now or before it was placed in the support. A first sealing member **190** is seated on the pivot tube **110** most proximate the interior surface of the spring case. The torsion spring with its spring tape **140** is seated in the receptacle **113** (best seen in FIG. **4**) with the outer spring end **116** in the open notch **119** (best seen in FIG. **6**). The plug **196** can be inserted now or after the tensioner's main body is assembled. Next, the spring bushing **198** is seated on the torsion spring with the outer ring's upper flange **208** seated against an exterior surface of the spring case. An annular pivot bushing **133** is either inserted into the pivot tube-receiving feature **132** of the arm **106** or is seated on the pivot tube. The pivot tube-receiving feature **132** is then seated over the pivot tube **110** and a second sealing member **192** is seated on the pivot tube **110** against the annular pivot bushing. The damper **170** is seated in keyed engagement with the arm **106** and then the support base **108** is seated on the pivot tube **110**. In this assembly process, the support base **108** is the last component of the main body **102** that is assembled. The pivot tube **110** is then fixedly connected to the support base **108**. The fixed connection can be produced by radially riveting the pivot tube to the support base or by another suitable fastening means. Fastening means can include any type of welding technique, including standard solder welding, spring welding, laser welding. Fastening means can include a bolt, or other fastener, adhesive, and other equivalents to any and all of the fastening means.
(23) With reference to FIGS. **2** and **4**, one benefit to the order of assembly of the tensioner **100** is that since the mounting bore **122***b* of the support base and the mounting bore **118** of the arm **117** of the spring case **111** are coplanar, see the plane line at datum A in FIG. **3**, the portion of the arm **117** defining the bore **118** can act as a registration stop and an alignment feature for the tooling used to press the support base **108** onto the pivot tube **110**.
(24) After the main body of the tensioner is assembled, the assembly can be flipped over and the pulleys **150** and **160** can be installed. Alternately, the pulley **150** can already be installed on the arm before it is seated on the pivot tube, and pulley **160** can already be installed on the support base before it is seated on the pivot tube.
(25) The belt tensioners discussed herein have numerous advantages, many of which have been already discussed above. Some additional advantages are provided in this paragraph. A flat wire spring is advantageous because it has a lower torque per degree of rotation and less degrees of variation than a round wire spring. A flat wire spring also has less resonance issues than a round

wire spring and the use of spring tape between the spring's coils can further reduce noise. Additionally, the flat wire spring reduces the tensioner's axial height (H), which can be advantageous in installing the tensioner in various motor configurations. The important advantage of the belt tensioner discussed herein is the reduction in the flexing or bending forces acting on the support base.
(26) It should be noted that the embodiments are not limited in their application or use to the details of construction and arrangement of parts and steps illustrated in the drawings and description. Features of the illustrative embodiments, constructions, and variants may be implemented or incorporated in other embodiments, constructions, variants, and modifications, and may be practiced or carried out in various ways. Furthermore, unless otherwise indicated, the terms and expressions employed herein have been chosen for the purpose of describing the illustrative embodiments of the present invention for the convenience of the reader and are not for the purpose of limiting the invention.
(27) Having described the invention in detail and by reference to preferred embodiments thereof, it will be apparent that modifications and variations are possible without departing from the scope of the invention which is defined in the appended claims.

Claims

1. A belt tensioner comprising: a spring case defining a receptacle for a torsion spring, the spring case comprising a first arm extending radially outward and axially rearward therefrom and terminating with a bore configured for fastening the spring case directly to an engine; a support base non-rotationally connected to the spring case by a pivot tube that defines a first axis of rotation, the support base having a pulley mount that defines a second axis of rotation and at least one bore configured for fastening the support base to the engine; a second arm having a first end coupled to the pivot tube for rotation about the first axis relative to the spring case and support base and having a second end that defines a pulley mount that defines a third axis of rotation; and a torsion spring having a first end operatively coupled to the support base and a second end operative coupled to the second arm, wherein the torsion spring biases the second arm in a belt engaging direction relative to the support base.
2. The belt tensioner of claim 1, wherein the pivot tube is positioned outside an area defined by the bores.
3. The belt tensioner of claim 1, wherein the torsion spring is a flat wire spring.
4. The belt tensioner of claim 1, wherein the first end of the spring is attached to the spring case.
5. The belt tensioner of claim 4, wherein the first end of the second arm includes a pivot tube-receiving feature, and the second end of the spring is operatively attached to the pivot tube-receiving feature.
6. The tensioner of claim 1, further comprising a first pulley operatively connected to the pulley mount of the support base for rotation about the second axis and a second pulley operatively connected to the second end of the second arm for rotation about the third axis.
7. The belt tensioner of claim 1, wherein the pivot tube has a knurled first end fixedly connected to the spring case and a knurled second end fixedly connected to the support base.
8. The belt tensioner of claim 1, further comprising a spring bushing seated between the second arm and the torsion spring, and a damper connected to the second arm for rotation therewith and in operational engagement with the support base.
9. The belt tensioner of claim 8, wherein the damper is keyed to the first end of the second arm for rotation with the second arm.
10. The belt tensioner of claim 8, wherein the spring bushing comprises a center ring and an outer ring connected to one another by a plurality of spokes.
11. The belt tensioner of claim 10, wherein the spring bushing includes an upper outermost axial

flange and a lower outermost axial flange, wherein the upper outermost axial flange is seated against an exterior surface of the spring case and the lower axial outermost flange is seated against an exterior surface of the second arm.

12. The belt tensioner of claim 1, wherein the spring case has an arm travel limiting feature defining a preselected number of degrees of rotation for the second arm and the second arm has a stop configured to engage the arm travel limiting feature.

13. A system comprising: a belt tensioner comprising: a spring case defining a receptacle for a torsion spring, the spring case comprising a first arm extending radially outward and axially rearward therefrom and terminating with a bore configured for fastening the spring case directly to an engine; a support base non-rotationally connected to the spring case by a pivot tube that defines a first axis of rotation, the support base having a pulley mount that defines a second axis of rotation and at least one bore configured for fastening the support base to the engine; a first pulley operatively seated on the pulley mount of the support base; a second arm having a first end coupled to the pivot tube for rotation about the first axis relative to the spring case and support base and having a second end that defines a pulley mount that defines a third axis of rotation; a second pulley operatively seated on the pulley mount of the second arm; a torsion spring having a first end operatively coupled to the support base and a second end operative coupled to the second arm, wherein the torsion spring biases the second arm in a belt engaging direction relative to the support base; and an endless belt engaging the first pulley and the second pulley.

14. The system of claim 13, wherein the pivot tube is positioned outside an area defined by the bores.

15. The system of claim 13, wherein the torsion spring is a flat wire spring.

16. The system of claim 13, wherein the first end of the spring is attached to the spring case, the first end of the second arm includes a pivot tube-receiving feature, and the second end of the spring is operatively attached to the pivot tube-receiving feature.

17. The system of claim 13, wherein a first end of the pivot tube is fixedly connected to the spring case and a second end of the pivot tube is fixedly connected to the support base.

18. The system of claim 13, further comprising a spring bushing seated between the second arm and the torsion spring, and a damper connected to the second arm for rotation therewith and in operational engagement with the support base.

19. The system of claim 18, wherein the damper is keyed to the first end of the second arm for rotation with the second arm.